

Team 1

Mr. Kiser

Engineering Service Learning

20 January 2026

Element D:

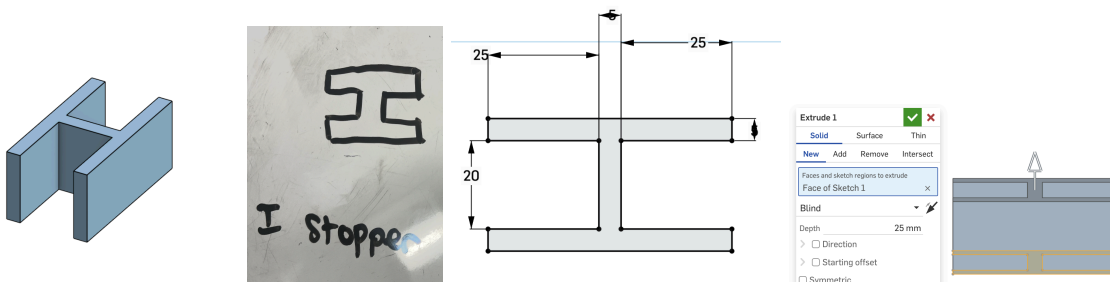
Criteria	Weight	Solution A (I-shaped stopper)		Solution B (Wedge)		Solution C (Clamp)		Solution D (Suction Cup)		Solution E (Foldable Lock)	
		Rating	Weight Score	Rating	Weight Score	Rating	Weight Score	Rating	Weight Score	Rating	Weight Score
Locking Ability	25%	2	0.5	4	1	5	1.25	2	0.5	4	1
TSA Compliant	20%	5	1	5	1	5	1	5	1	5	1
Lock in < 5 seconds	15%	5	0.75	5	0.75	1	0.15	5	0.75	4.5	0.675
Size	15%	4	0.6	4	0.6	2	0.3	5	0.75	2	0.3
Durability	10%	4	0.4	3	0.3	4	0.4	2	0.2	3	0.3
Additional Function*	10%	4	0.4	4	0.4	1	0.1	3	0.3	3	0.3
Material usage	5%	5	0.25	5	0.25	5	0.25	3	0.15	3	0.15
SUM	100%		3.9		4.3		3.45		3.65		3.725
*This includes being able to use one hand and less than five pounds to operate the lock											

Weight Justification: We considered Locking Ability the highest priority because the lock needs to be able to lock a bathroom door or else it is unusable. TSA compliant has the second highest weight because the primary stakeholder expressed how he would be using this lock on a plane or at an airport, so the lock must be TSA compliant or he cannot take the lock with him. Lock in less than five seconds and size are the next highest priority based on primary stakeholder data. He expressed how he would greatly appreciate having these features to the lock. Durability and additional functions are next because they are important to the stakeholder, but if they are not met, they wouldn't be a fundamental design flaw. Material usage has the lowest weight because a lock can still have all the design features that the stakeholder wanted, but it may cause it to be more expensive or harder to get.

Design Idea #1:

Date: 1/16/26

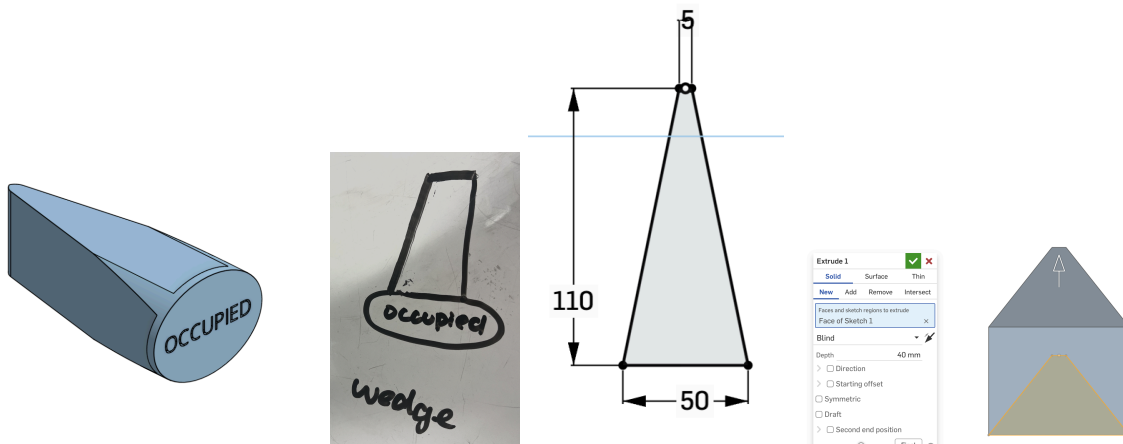
Description: This stopper would be in the shape of an I. The sides of the I would go around the door, and it would rest on the unfunctional lock. This would be completely 3D printed; therefore, the cost would be nothing.



Design Idea #2:

Date: 1/16/26

Description: This lock would lock into the gap of the bathroom stall. Because of friction, it would stay still. We also wanted to add an occupied sign on the front for added visibility, but this could be added to the other designs as well. The idea of this design is that unlike the I stopper, if there is no lock assembly at all, the lock would still be able to stop the door and keep it shut. The plastic and tape are the only materials and would be found in class, so there would be no cost.

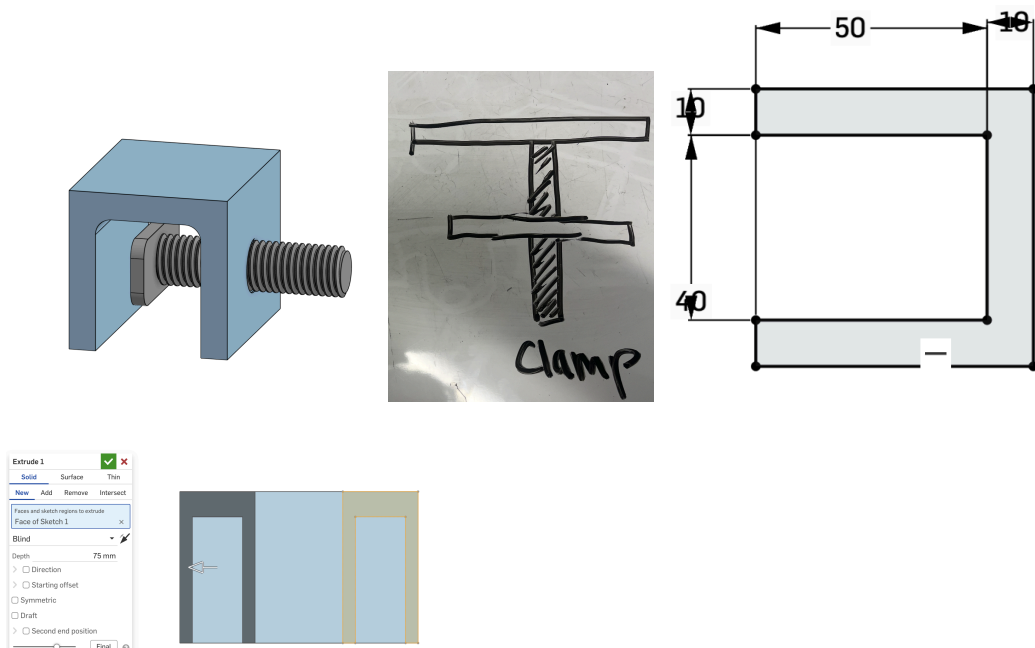


Design Idea #3:

Date: 1/16/26

Description: This lock would have two flat faces with a screw like mechanism that can clamp the faces of the door together. The clamp would be a very effective lock, but would not fit the other criteria such as taking less than five seconds, or being only from the outside of the stall.

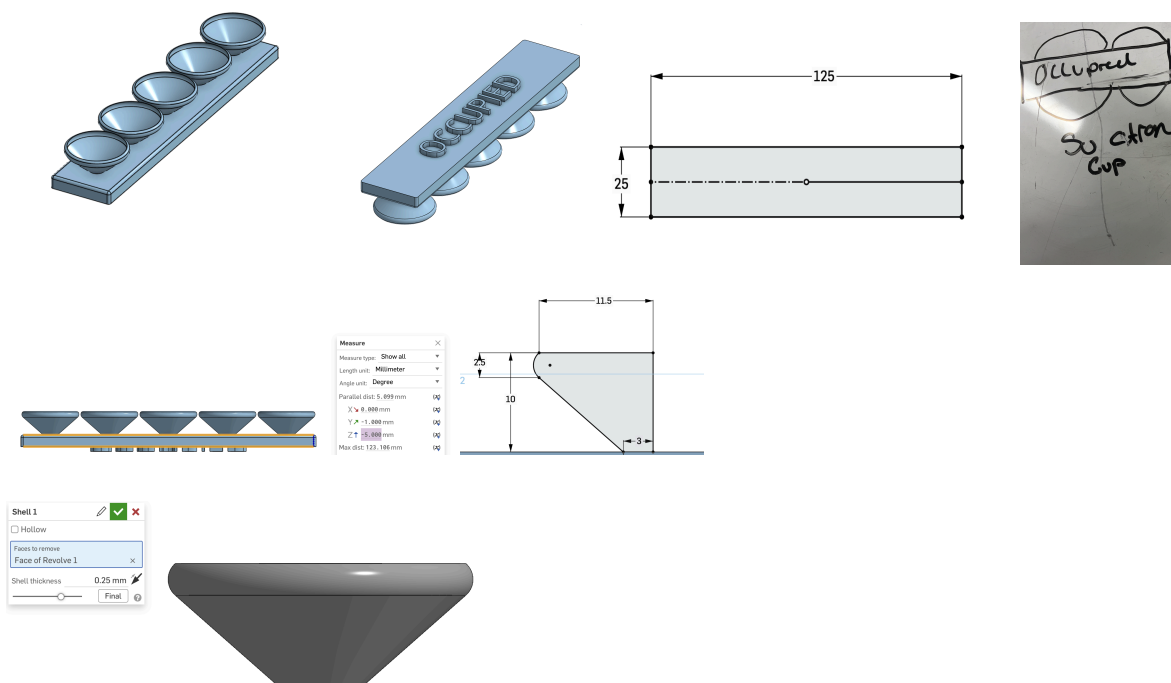
However, we thought that it would be very effective at keeping the door shut. This would include a screw, a nut, and two wooden planks that would be found in class, so there would be no cost.



Design Idea #4:

Date: 1/16/26

Description: This idea would use a line suction cups attached to a plank of wood or a 3-D printed rectangular prism to hold the two sides of the door together. The idea was that the suction cups would grip easily to the bathroom door as well as being easy to remove-this could be a pro or a con to this design. This would make for a really compact design, however we have evidence to suggest that it would not be very durable and the materials could be hard to find. If we had the materials they would be 5 suction cups, one wooden plank, and hot glue, they would be found in class and have no cost.

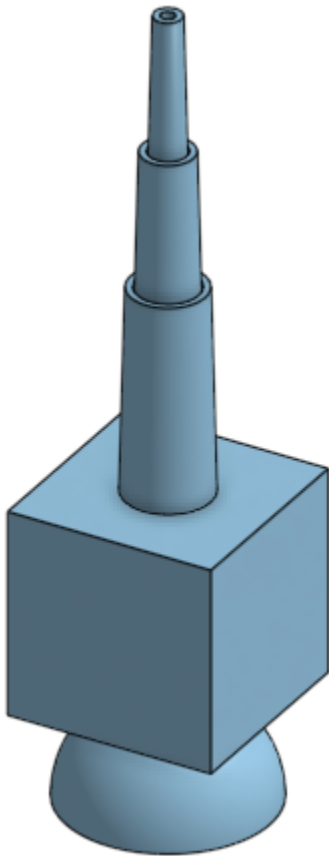


Design Idea #5

Date: 1/16/26

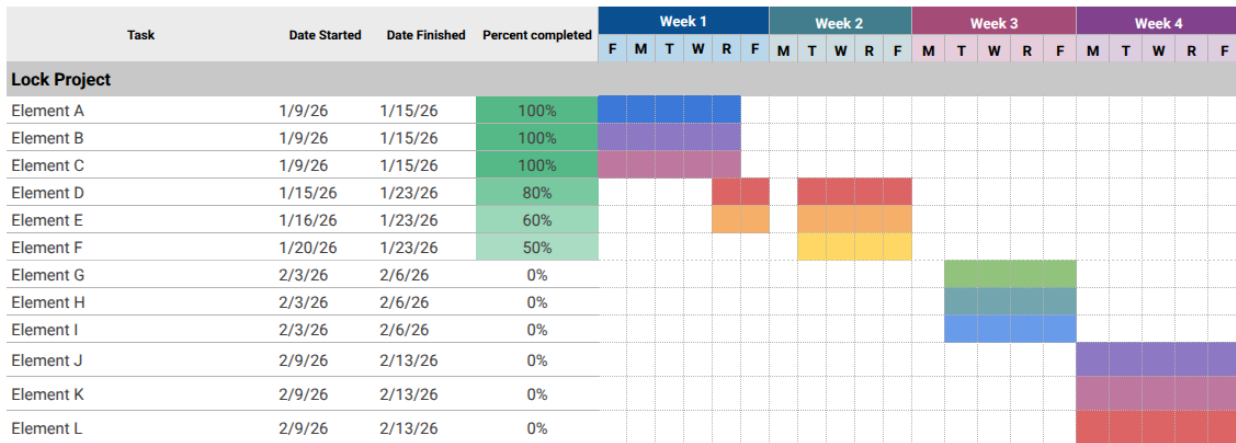
Description: This design would basically be a foldable block and rod that would sit on the floor, and the rod would poke up. There would be a suction cup on the problem to allow it to stay, and

it would allow for easy use. Unfortunately, while brainstorming it became much more difficult to make something that had a long enough stick and block to truly prevent movement with the given materials. However, if the right materials were found, take a small, wooden rectangular prism with a small hole in the middle. Attach a strong suction cup to the bottom of the prism and use strong glue to attach a long retractable metal pole to the prism. This would consist of a retractable metal pole, a suction cup, a wooden block, and hot glue sticks. They would be found in class, and they would have no cost.



Gantt Chart:

Lock Gantt Chart



Manufacturing Method for All Designs:

1. Export the design from Onshape as an stl file
2. Open the file in Bambu Lab Studio
3. Adjust the 3D printer settings as desired (such as color and infill)
4. Print the model and wait for the lock design to print (Must have a Bambu Lab printer with AMS)